Request for Proposal

The Society of Cedarcrest Engineers (hereafter referred to as SCE) is seeking proposals for a Golf Putter Tester (GPT). All proposals must include drawings of the GPT, testing plans and results, scale models, and a physical assembly. All Golf Putter Testers will be tested on December 09, 2026. All written proposals (including drawings) will be due to the association on or before January 11, 2027. The design teams submitting proposals for consideration will present their designs between January 04-20, 2027.

Design Teams

There may be no less than two and no more than three persons per team. Each design team must have a team name, logo (drawn electronically), mission statement, motto, and song (no greater than 30 seconds). The team logo must appear on all drawings and on the first page of the requirements document. The team name, mission statement, motto, and song must be presented during the oral presentation.

Requirements

The Golf Putter Tester must meet certain operational and design requirements. Failure to meet the design requirements will make the GPT eligible for only 10 of the 20 performance points possible. Systems that meet the design requirements will be eligible for a maximum of 30 points (up to 10 extra points, as long as the total points for the entire project is not exceeded) depending on how the systems perform during testing.

Design Requirements: The Golf Putter Tester must:

1. be digitally designed and assembled on SolidWorks,
2. have a footprint no larger than 18" x 18" when assembled,
3. have a minimum of 6 unique parts (minimum of 2 designed per person) and may have up to three other commercial parts as part of the design (but these parts do not count as any of the 6 unique parts),
4. have at least one 3D printed part with a total 3D printed cost under $10 as shown by the 3D printer software,
5. be able to hold a golf putter and make it strike a golf ball so that it rolls on the ground,
6. be designed, constructed, tested, drawn, and documented by the design team.

Performance Requirements: RT must:

1. putt into a golf hole from 5 feet, 10 feet, and 15 feet,
2. putt 6 golf balls into the hole with 8 attempts.

Points:

Has six unique parts.	3 points
Meets footprint size requirement.	5 points
Meets 3D printed part cost requirement (a penalty of half a point for every $0.50 over the cost requirement will be assessed).	(Max 5 points subtracted)
Meets all other design and performance requirements.	2 points
Putts 3 golf balls into the hole from the three different distances (5', 10', 15')	5 points
Putts 6 of 8 golf balls into the hole	5 points

Bonus (Extra) Points:

Putts 6 of 6 golf balls into the hole with 2 from each of the three different distances	2 points
Putts 8 of 8 golf balls into the hole with a minimum of 2 each from each of the three different distances.	5 points
Putts 8 of 8 golf balls into the hole with 2 at 5 feet, 2 at 10 feet, and 4 at 15 feet.	5 points

If time allows, teams will get a second chance to try their putting of 8 golf balls. The second chance starts from zero.

A maximum of 30 performance points can be received. All teams submitting a prototype system for testing will receive a minimum of 5 points. Teams submitting projects made from kits will receive no points. Teams submitting projects not designed, built, and tested by the team will receive no points.

Written Requirements

Each design must be detailed by a requirements document which must follow the requirements document boilerplate (pr_rqtsdoc.doc). The boilerplate document should be completed according to the directions given in the paragraphs. The formatting of the document should not be changed. All the sections of the document must be complete and drawings must be included. The document should be completed as a third-person narrative. References to the author(s) should also be made in the third-person.

Drawing Standards

Drawings must follow the drawing standards as defined in Drawing Requirements Manual, Ninth Edition. Particularly, the drawing formats found in section 6 and the drawing and part numbering systems found in section 7 must be followed. All drawings will have a standard title block (with the design group identifiers) as found in section 6.

Test Plans and Results

As stated above, testing must be documented and included in the written requirements. The test plans must have a clearly stated objective and detailed procedures that will meet that objective. As part of the test results, make observations and write them in a test notebook. All of the notes, including observations from the tests must be included in the test results. A bound, page-numbered notebook is acceptable for testing

(it is required by the U.S. Patent office) and this may be included in its entirety as part of Appendix A.

Oral Presentations

Design teams will present their products after the tests on the dates given above. All presentations must include models, drawings, and other visual aids. Each member of the design teams must be part of the presentation. The presentations must be from 12 to 25 minutes (for the paired teams – essentially 6 minutes per team member minimum). The presentation requirements and point distribution is detailed in the file Rubric-Presentation.docx (available online at: LearningWithMiyoshi/_CommonDocuments/Rubric-Presentation.docx at main · MichaelTMiyoshi/LearningWithMiyoshi).

Scoring

There will be a total of 120 points possible for each person on the design team. The points will be awarded as follows:

20 pts.	Meets Timeline Objectives
20 pts.	Meets Requirements
20 pts.	Requirements Document
20 pts.	Drawings and Models
20 pts.	Appendices A and B (Test Plans and Results, Cost)
20 pts.	Oral Presentation

Projects must meet the specifications in each category. Oral presentation requirements are found in the document Rubric-Presentation.docx. There are five timeline objectives that must be met. Each objective met (on time) will receive up to 4 points. The five objectives are as follows: 1. form groups, 2. create team name, logo, and mission statement, 3. design tests, 4. create initial drawings or create test parts, 5. create draft of requirements document and create visual aids for oral presentation.

The total out of one-hundred and twenty points will be awarded for the project and then multiplied by the number of team members. The team members will then be allowed to give points to the members as they have earned them. The group must all be in agreement as to the distribution of points before individual grades are given. (Timeline requirements and presentation points are excluded in the redistribution of points.) An individual's signature must be given to show agreement on the point distribution. Any team that cannot come to agreement on the point distribution will go to arbitration by the head of the Society of Cedarcrest Engineers (Mr. Miyoshi). Note that any individual in the group may receive a minimum of 20 points and a maximum of 120 points regardless of the total number of points (including bonus points) earned by the group.